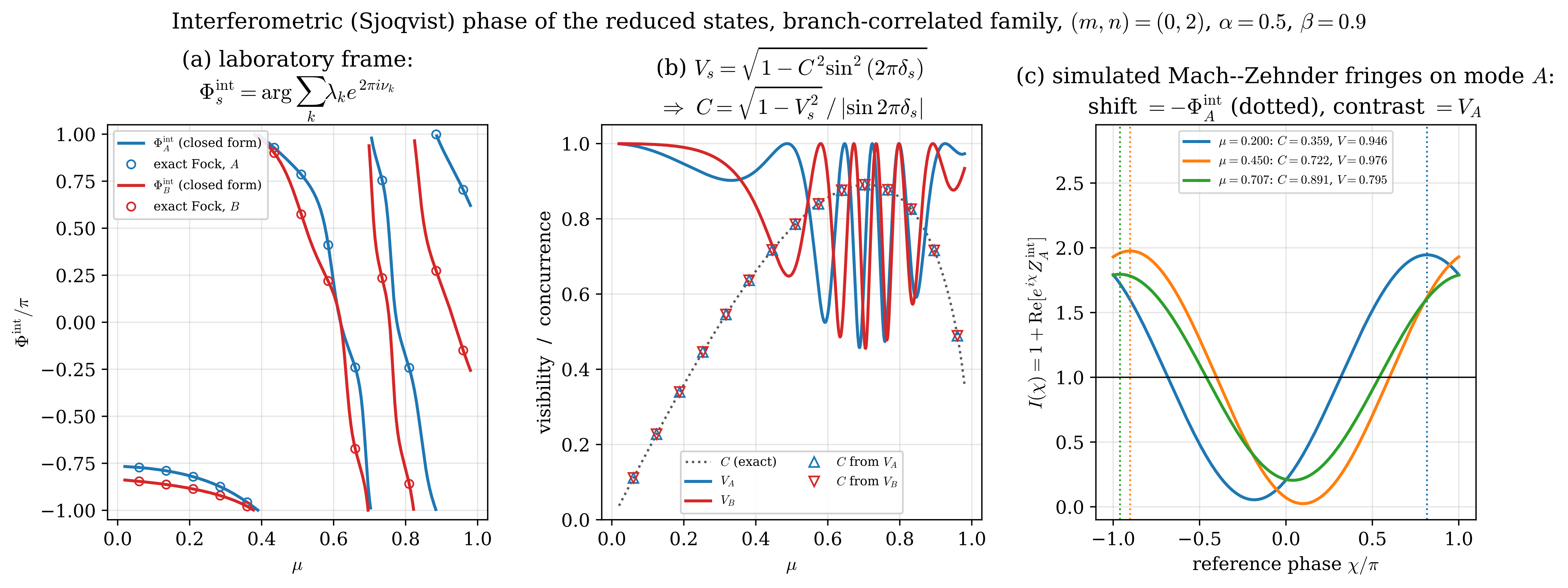

Interferometric (Sjoqvist) phase of the reduced states, branch-correlated family, $(m,n)=(0,2)$, $\alpha=0.5$, $\beta=0.9$
(a) laboratory frame:
$\Phi_s^{\rm int}=\arg\sum_k \lambda_k e^{2\pi i\nu_k}$
$\Phi_A^{\rm int}$ (closed form)
exact Fock, A
$\Phi_B^{\rm int}$ (closed form)
exact Fock, B
$\Phi^{\rm int}/\pi$
μ
(b) $V_s=\sqrt{1-C^2\sin^2(2\pi\delta_s)}$
$\Rightarrow\ C=\sqrt{1-V_s^2}\ /\ |\sin 2\pi\delta_s|$
visibility / concurrence
C (exact)
V_A
V_B
C from V_A
C from V_B
μ
(c) simulated Mach--Zehnder fringes on mode A:
shift $=-\Phi_A^{\rm int}$ (dotted), contrast $=V_A$
$\mu=0.200$: $C=0.359$, $V=0.946$
$\mu=0.450$: $C=0.722$, $V=0.976$
$\mu=0.707$: $C=0.891$, $V=0.795$
$I(\chi)=1+\mathrm{Re}[e^{i\chi}Z_A^{\rm int}]$
reference phase χ/π